

Unifying compound specific phonological effects in Malayalam - a case for moraic affixation

Katie McCann (Universität Leipzig),

Jochen Trommer (Universität Leipzig)

Abstract: This article tackles the morpho-phonological complexities of compound-specific phonological effects in Malayalam. We argue that all effects of lengthening and their interaction with epenthesis underlie a single source, namely a linking element realised as a moraic affix. The realisation of this moraic affix is sensitive to lexical, morphosyntactic structural and phonological factors. Despite the non-phonological influences on the moraic affix’s realisation, we contend that the phonology neither needs to reference lexical information nor is it necessary to invoke non-restrictive interleaving of morphology and phonology to account for the structural influence. Instead, three assumptions determine the distinct surface patterns of the moraic affix. First, the linking element has two lexically conditioned allomorphs which are both realised by a floating mora but vary in their underlying strength, modeled as gradient symbolic representations. Second, lengthening is sensitive to prosodic structure, which differs between compound types due to morphosyntactic structural differences. Third, general phonological well-formedness interacts with how the floating mora is realised. These three assumptions allow all configurations of lengthening and insertion at the compound boundary to be unified by a single instance of a linking element. (word count: 187)

Keywords: moraic affixation, moraic theory, linking elements, gemination, gradience, compound phonology

1 Introduction

At a compound boundary in Malayalam, different effects of lengthening and epenthesis occur. We identify six configurations that lead to distinct surface phonological effects. In this article, we show that all effects originate in a moraic affix, more precisely a moraic linking element. We begin by describing the phonological effects at a compound boundary in Malayalam, as they are described by Mohanan (1982, 1986). These include either lengthening or epenthesis. In example (1a), the initial plosive of the second noun in the compound lengthens to a geminate. In (1b), it is the final consonant /t/ of the first noun in the compound that geminates. In

(1c), the final vowel of the first noun in a compound lengthens. In (1d) a schwa is epenthesised between the two underlyingly adjacent plosives at the compound boundary and no lengthening is observed.

- (1) a. [a:na^{kk}u^{ti}ra]
a:na-ku^{ti}ra
elephant-horse_[DRAVIDIAN]
‘horse that is like an elephant’ (subcompound)(p.90, Mohanan 1986¹)
- b. [ka:^{tt}a:na]
ka:t-a:na
forest-elephant
‘untamed elephant’ (subcompound)(p.91)
- c. [tar^{a:}ka:^{nt}an]
tara-ka:^{nt}an
Tara-husband_[SANSKRIT]
‘Tara’s husband’ (subcompound)(p.36, Mohanan 1982)
- d. [camp^əpa:^{tr}am]
camp-pa:^{tr}am
copper-vessel_[DRAVIDIAN]
‘copper vessel’(subcompound)(p.85)

In each example in (1), we observe only one instantiation of lengthening or epenthesis. Take examples (1a,b), where it is only the consonant at the compound boundary that is geminated, even though there is also a vowel that could be lengthened. Conversely, in the example (1c), there is only the effect of a lengthened vowel and no gemination occurs. In (1d) neither consonant at the compound boundary is geminated, only a schwa is epenthesised. We conclude that all compound-specific epenthesis and lengthening stand in complementary distribution. This holds across all examples presented by Mohanan (1982, 1986), with one systematic exception. A stem-final /t/ or /r/² is always geminated at a compound boundary, even if the resulting consonant cluster necessitates schwa epenthesis, see example (2). Compare this to example (1d), where there is no stem final /t/ or /r/, a schwa is epenthesised but no gemination surfaces.

- (2) [a:^{tt}^ətozu^{tt}ə]
a:t-tozu^{tt}
sheep-stable
‘sheep shed’

¹All page numbers in this and future example refer to Mohanan (1986) if not otherwise indicated.

²When /r/ is geminated it is pronounced as [tt].

Given that all compound-specific phonological effects stand in complementary distribution, with one exception, it stands to reason that the compound-specific effects arise from the same source. In this article, we establish the hypothesis that all effects of lengthening and its interaction with epenthesis in (1) and (2) arise through the affixation of a moraic linking element. This moraic linking element causes different surface realisations due to three factors in the morphophonology. Purely phonological factors influence the realisation of the moraic linking element. If the final consonant of the first noun is either an /t/ or an /r/, then this consonant will always be geminated. Even if there exists another target for gemination in the initial consonant of the second stem, as in example (2), gemination of /t/ or /r/ is always preferred. When there is no /t/ or /r/ and instead two plosive consonants end up adjacent at the compound boundary, the illicit consonant cluster is repaired by schwa epenthesis which bleeds gemination.

Interestingly, the choice between consonant gemination and vowel lengthening cannot be phonological. Compare the example in (1a) to that in (1c). In both examples, the initial noun is vowel-final and the second noun is consonant-initial. Here, as pointed out by Mohanan (1986), the compound that has a second noun of [DRAVIDIAN] origin surfaces with the consonant geminate, as in (1a). The compound that has a second noun of [SANSKRIT] origin surfaces with a long vowel, as in (1c). Thus, the determining factor between consonant gemination and vowel lengthening is lexical. Further, in [SANSKRIT] stems, it is only the final vowel of the first stem that can lengthen. If there is no final vowel then no lengthening effect is observed, even if a consonant could be geminated without leading to phonological ill-formedness, see example (3a). In the same vein, if the second stem has [DRAVIDIAN] origin and there is no stem-initial plosive and no stem-final /t/ or /r/ consonant at the compound boundary, there is no lengthening effect observed, see example (3b).

- (3) a. [kuɭirkaːttə] *[kuɭirkkaːttə]
 kuɭir-kaːtt
 cold-breeze_[SANSKRIT?]
 ‘cool breeze’ (subcompound) (p.85)
- b. [wirakaɽuppə] *[wirakkaɽuppə] *[wirakaɽuppə]
 wirak-aɽupp
 firewood-LE-stove
 ‘stove that uses firewood’ (subcompound) (p.91)

Aside from phonological and lexical factors that determine the type of phonological effect at the compound boundary, a third factor also influences the application of consonant gemination. The compounds in (1), (2) and (3) all have a subordinate compound structure; they are all so-called subcompounds or tatpuruṣas consisting of a modifier+head structure. Malayalam also forms coordinative compounds or dvandva, see Wälchli (2005) and Ralli (2019) for a cross-linguistic overview of coordinative compounds. In so-called co-compounds, no geminate consonant surfaces, even if the second stem has [DRAVIDIAN] origin and is plosive-

initial. Compare the co-compound example in (4a) to the subcompound in the example (4b). Gemination is blocked in the co-compounds but not in the subcompound.

- (4) a. [petti[̥]p[̥]atta:jaɳɳalə]
petti-patta:jam-kal
box-grainbin_[DRAVIDIAN]-PL
‘boxes and grainbins’ (co-compound)(p.90)
- b. [petti[̥]pp[̥]atta:jaɳɳalə]
petti-patta:jam-kal
box-grainbin_[DRAVIDIAN]-PL
‘grainbins used as boxes’ (subcompound)(p.90)

Vowel lengthening is not restricted by the coordinative compound structure. In the example (5), the co-compound surfaces with a lengthened vowel in the first noun stem.

- (5) [ba:lik[̥]a:ba:lanma:rə]
ba:lika-ba:lan-ma:r
girl-boy_[SANSKRIT]-PL
‘girls and boys’ (co-compound) (p.44, Mohanan 1982)

To sum up the data, we observe four different effects of lengthening or epenthesis at the compound boundary in Malayalam: initial gemination of a plosive, final vowel lengthening, ə-epenthesis between two obstruents and final t/r gemination. Three factors determine which process will surface. These are the phonological environment, the lexical environment (the origin of the second stem) and the structural environment (the type of compound). Aside from final t/r gemination and ə-epenthesis, all effects stand in complementary distribution. Table 1 summarises the observed effects and their determining factors. Taken together, there are six possible configurations that lead to six different phonological effects at the compound boundary (abbreviations used in the table: C = consonant, P = plosive, S = sonorant, V = vowel).

environment	phonological lexical structural subcomp	V-P/S-P [DRAVID] any	V-C [SANSKRIT]	any [DRAVID] co-comp	P-P any any	t-P/r-P [DRAVID] subcomp	t-V/r-V [DRAVID] subcomp
phon. effect	initial gemination	VP:/SP:	–	–	–	–	–
	vowel lengthening	–	V:C	–	–	–	–
	ə-epenthesis	–	–	–	PəP	t:əP/t:əP	–
	final t/r gemination	–	–	–	–	t:əP/t:əP	t:V/t:V

Table 1: Summary of all lengthening/epenthesis effects at the compound boundary

In this paper, we tackle all six configurations presented in the table above. Our analysis unifies all lengthening effects to arise from a single underlying source. This source is a compound internal linking element, which is expounded by a moraic affix. (word count (excluding examples): 932)

1.1 Malayalam in Lexical Morphology and Phonology

The data presented in this paper is introduced and analysed by Mohanan (1982, 1986) in the framework of Lexical Morphology and Phonology (LMP). This section provides a short overview of his analysis. Mohanan (1986) claims that compound-specific phonological effects of Malayalam are instantiated by multiple separate phonological rules, which stand in disjunctive relation only because they refer to different lexical and morphological information. The three distinct phonological rules he proposes are presented in (6). The first for stem-initial consonant gemination in (6a), the second for vowel lengthening in (6b) and the third for final gemination of *t* and *r* in (6c).

- (6) a. *Stem-initial consonant gemination in Mohanan (1986: 90):*

$$\begin{array}{c} C \rightarrow CC / X \quad || \text{ — } \\ | \quad \vee \quad | \\ \left[\begin{array}{c} +\text{son} \\ -\text{nas} \end{array} \right] \left[\begin{array}{c} -\text{son} \\ +\text{Dravid} \end{array} \right] \end{array}$$

- b. *Vowel lengthening Mohanan (1986: 100):*

$$\begin{array}{c} N \rightarrow N / \text{ — } || \\ | \quad \swarrow \quad \searrow \\ \quad X \quad X \\ \quad \searrow \quad \swarrow \\ \quad \quad X \end{array} \left[\begin{array}{c} +\text{Sanskrit} \end{array} \right]$$

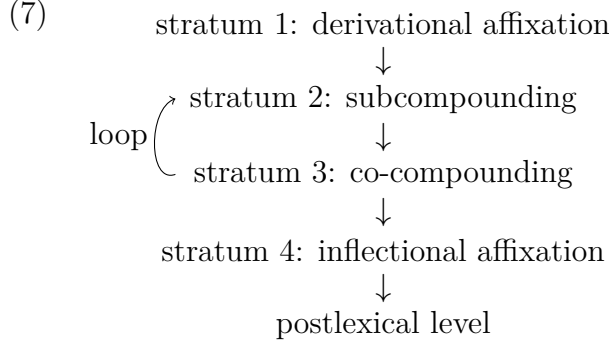
- c. *t and r gemination in Mohanan (1986: 92)*

$$\begin{array}{c} C \rightarrow CC / X \quad || \\ | \quad \vee \quad | \\ \quad \quad \left\{ \begin{array}{c} t \\ r \end{array} \right\} \end{array}$$

The phonological rules in (6) must have access to lexical information to determine their applicability.

Interestingly, the rules in (6) do not make reference to a difference between application in subcompounds and co-compounds. The rule in (6a) does not refer to the morphosyntactic structure of the compound. This is unexpected given that consonant gemination only applies in subcompounds and not in co-compounds. To arrive at the phonological distinction between the compound types, Mohanan assumes that subcompounds and co-compounds are built at distinct lexical strata. Thus, the non-application of consonant gemination in co-compounds is derived in Mohanan's (1986) framework by assuming that the rule of consonant gemination in (6a) is only part of the phonology of the subcompounding stratum. The stratal

architecture he proposes is presented in (7). Crucial to his analysis is that subcompounds and co-compounds are built at different strata because they undergo different phonologies. However, morphologically the sub- and co-compound strata interact. Subcompounds can be embedded in co-compounds and co-compounds can be embedded in subcompounds. To derive these morphological interactions while maintaining the stratal phonological distinction between compounds, Mohanan posits a stratal loop which loops from the co-compounding stratum back to the subcompounding stratum.



The stratal loop has been criticised by Sproat (1985), Christdas (1986), and Inkelas (1989) shortly after its introduction for weakening the theory of Lexical Morphology and Phonology. Crucially, the introduction of the loop abandons strict level ordering (Kiparsky 1982). We propose in this article that there is no need to resort to a loop and therefore no need to abandon strict level ordering. The three rules in (6) can be collapsed into a single linking element. Thus, subcompounds and co-compounds are morphologically built at the same stratum where they undergo the same phonological grammar. The phonological differences arise through the morpho-phonological restrictions on the linking element. Three factors influence the realisation of the linking element. First, the lexical environment determines the choice between two allomorphs of the linking element. These allomorphs are both realised by a floating mora but are differentiated representationally by their underlying strength, modelled in gradient symbolic representations (Smolensky & Goldrick 2016, Rosen 2016, Zimmermann 2019). The difference in strength causes either vowel lengthening or consonant gemination in the phonological component. This crucially means that the phonology need not access lexical information, because allomorphy is handled by the morphological component. Second, the morphosyntactic structure determines different prosodic parses between subcompounds and co-compounds, to which the consonant geminates are sensitive. This derives the non-realisation of consonant gemination in co-compounds without resorting to excess strata and a loop. Third, the phonological environment determines whether the moraic linking element causes lengthening or schwa epenthesis or both. The rest of this article presents all three factors in turn. Section 2 begins with the introduction of the moraic linking element and its allomorphic variants. In this section, we assert that strong moras prefer heading vocalic elements, while weak moras can head consonantal elements. In section 2.1 we embed the differentiation between strong and weak moras into the gen-

eral phonological grammar of Malayalam and show that the difference in strength aligns with phonological properties in long vowels compared to geminate consonants concerning stress and minimal word effects. In section 3, we further the argument of moraic affixation by extending the analysis to the phonological influences on compound-specific phonology. We begin in section 3.1 by asserting that the structural restriction on gemination in co-compounds is reflected as a prosodic restriction on gemination. In section 3.2, moraic affixation is shown to derive the bleeding effect of schwa epenthesis on stem-initial gemination. In addition, we show that moraic affixation also derives a feeding effect of stem-initial gemination on final degemination. Finally, in section 3.3, we extend the analysis to final /t/ and /r/ gemination. Section 4 provides a discussion and concludes that the phonological grammar of Malayalam neither requires reference to lexical information in the phonology, nor excess strata, nor a stratal loop.

2 Compound specific phonology is moraic affixation

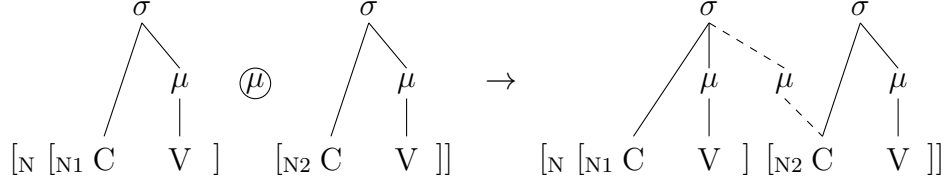
In this section, we establish our main claim, that all compound-specific phonological effects in Malayalam can be traced back to the phonological realisation of a morphological linking element. We postulate that a linking element is realised by a moraic affix at the compound boundary in Malayalam. A linking element is defined as phonological material at the boundary of a compound that resembles morphological material, despite being (prototypically) meaningless (Kürschner & Szczepaniak 2013). Linking elements are common in many Indo-European languages, for example in German (Fuhrhop 1996, Aronoff & Fuhrhop 2002). Take the example in (8) of a German compound. The linking element appears as an [s] between the two nouns of the compound.

- (8) *German linking element* (LE)
Hochzeit-s-torte
wedding-LE-cake
‘wedding cake’

Recent literature has argued that linking elements are not limited to segmental elements but may also consist of phonological features. Labrune (2014) shows that Basque *lotura* are analysable as featural linking elements, as are Japanese *rendaku* (Labrune 1999, Itô & Mester 2003). We follow Labrune (2014) in treating Malayalam compound-specific lengthening as a reflex of a featural linking element. More precisely, the feature of the linking element is length, represented as a floating moraic affix. This floating mora is affixed between two nouns of a compound and integrates into the syllable structure resulting in either a lengthened vowel, a geminate consonant or in certain phonological environments an epenthetic schwa. Thus, Malayalam constitutes a case of moraic affixation (Davis & Ueda 2002, Kirchner 2009, Trommer & Zimmermann 2014) in the form of a linking element. As shown schematically in (9) one result of the integrated moraic linking element is consonant gemination. In (9) the linking element is realised between the two

nominal elements N_1 and N_2 as a floating mora. In the output, the mora causes consonant gemination by associating to the initial consonant of N_2 thereby forming the coda of the final syllable in N_1 and the onset of the first syllable in N_2 , following Hayes (1989) theory of gemination.

(9) *Floating moraic linking element integrated to produce a consonant geminate*



As is typical for linking elements, the floating moraic linking element in Malayalam is semantically empty. We do not claim that the linking element realises grammatical material but follow proposals by Fuhrhop (1996), Ralli (2008), and Labrune (2014) in assuming that the purpose of the linking element is to signal compounding.

Thus far, we have established that lengthening at a compound boundary in Malayalam is a phonological reflex of a morphological linking element. However, we have not determined how the different phonological effects come to be. As presented in the previous section, three factors influence whether and how the mora is realised; lexical, structural and phonological. In this section, we develop the lexical influence on moraic affixation of the linking element. Remember, that whether a vowel lengthens or a consonant geminates is determined on purely lexical grounds, namely whether the second stem of the compound has Dravidian origin, see example (10a) repeated from (1a), or Sanskrit origin, see example (10b) repeated from (1c).

- (10) a. $[a:na\boxed{kk}u\grave{t}ira]$
 a:na-kuṭira
 elephant-horse_[DRAVIDIAN]
 ‘horse that is like an elephant’ (subcompound)(p.90, Mohanan 1986³)
- b. $[tar\boxed{a:}ka:n\grave{t}an]$
 tara-ka:nṭan
 Tara-husband_[SANSKRIT]
 ‘Tara’s husband’ (subcompound)(p.36, Mohanan 1982)

Thus, the difference between vowel lengthening and consonant gemination must originate in the morphology which has access to lexical information. We argue that the linking element has two allomorphic variants which are lexically determined. One allomorphic variant is realised as a strong floating mora (μ_1) which causes vowel lengthening in the phonological component. The other allomorphic variant is realised as a weaker floating mora ($\mu_{0.5}$) which causes consonant gemination in the phonological component. Thus, we argue that the different

phonological effects of lengthening arise through a difference in strength of the linking element allomorphs. We model the difference in strength using gradient symbolic representations (GSR) (Smolensky & Goldrick 2016, Rosen 2016, Zimmermann 2019). GSR introduces the notion that phonological elements are not categorically but gradiently active. This notion has been shown to derive exceptional phonological patterns (Zimmermann 2019) and phonological variability (Rosen 2016). Indeed, the two allomorphic variants of the linking element in Malayalam are reminiscent yet not parallel to Rosen’s (2016) analysis of Japanese voicing effects in compounds. He shows that variation in voicing of the initial consonant of the second noun in a compound arises from variation in strength of a floating [+vc] feature. Our analysis differs from Rosen’s (2016) in that we assume two distinct allomorphs of a linking element, whereas Rosen assumes that the [+vc] feature is part of the first noun in a compound and the initial consonant of the second noun is also be gradiently active. Our implementation of GSR into Malayalam as the activity of the floating moraic linking element suffices to derive the variable behaviour concerning vowel lengthening and consonant gemination. The two allomorphs of the linking element are shown in (11).

- (11) a. $(\mu_1) \leftrightarrow \text{LE} / \text{N}_1 \text{ — } \text{N}_{2[\text{SANSKRIT}]}$
 b. $(\mu_{0.5}) \leftrightarrow \text{LE} / \text{N}_1 \text{ — } \text{N}_{2[\text{DRAVIDIAN}]}$

The key insight behind deriving the variable behaviour of the linking elements is gradient constraint violation. To illustrate how this works, take the constraint MAX. MAX is violated by any input material that is deleted in the output. In GSR, MAX is violated to a higher degree if the deleted material has a high activity in the input and to a lower degree if the deleted material has a low activity in the input. Gradient violations are computed within Harmonic Grammar (Legendre et al. 1990, Potts et al. 2010) where constraints are not ranked with respect to each other but weighted. GSR in combination with Harmonic Grammar computes the harmony score of an output candidate by determining the gradient violations which are multiplied by the weight of the constraint that they violate. All candidate harmony scores are compared and the candidate with the harmony score closest to 0 wins. Tableau (12) demonstrates a derivation using GSR in Harmonic Grammar. The input consists of two consonants with different activities. /p_{0.2}/ has a low activity, /t₁/ is fully activated. The grammar consists of two constraints, *CC with a weight of 20 and MAX with a lower weight of 10. The constraint *CC is fully violated by the faithful output in (12a) due to the consonant cluster. There are two ways of resolving the consonant cluster. The first is to delete /t₁/ as shown in output candidate (12b). This incurs a full violation of MAX because the deleted segment is fully activated. This violation is multiplied with the constraint weight yielding an overall harmony score of -10. The alternative is to delete the weak consonant /p_{0.2}/ shown by output candidate (12c). Deletion of /p_{0.2}/ also violates MAX but to a lower degree due to the low activity of the segment. MAX in candidate (12c) is violated gradiently by -0.2 yielding the overall most harmonic score of the tableau, -2.

(12) Toy derivation of /pt/ → [t]

p _{0.2} t ₁		*CC 20	MAX 10	$\mathcal{H}$
a.	p _{0.2} t ₁	-1		-20
b.	p _{0.2}		-1	-10
 c.	t ₁		-0.2	-2

In the original proposal of GSR, GEN is restricted to allow only full activation (strength = 1) in the output (Smolensky & Goldrick 2016, Rosen 2016). Thus, an output segment with strength 0.2, as in output (12a) and (12b) is not created by GEN. Instead, the segment /p/ must be strengthened by 0.8 which incurs gradient violations of the DEP constraint. We depart from this assumption and follow recent work by Zimmermann (2018, 2019) which assumes that GEN can create gradiently active phonological elements in the output, as presented in the tableau (12) above.

Let us return to the case of Malayalam linking elements. We claim that a less activated floating mora, a weak mora, will integrate into the syllable structure and associate with a consonant which yields consonant gemination, whereas a fully activated floating mora integrates into the syllable structure by associating with a vowel which leads to vowel lengthening. To derive this difference, the following two constraints are assumed. The first constraint is $\boxed{*V^{\mu 0.5}}$ defined in (13a). This constraint is gradiently violated by any vowel that is dominated by a mora with an activity lower than 1. The second constraint is $\boxed{*C^{\mu 1}}$ defined in (13b). This constraint is gradiently violated by a consonant that is dominated by a mora stronger than 0.5. Both constraints receive the same weight of 10.

- (13) a. $\boxed{*V^{\mu 0.5}}$: Assign a violation x to every vowel that is dominated by a mora of strength y such that $x=1-y$. weight=10
- b. $\boxed{*C^{\mu 1}}$: Assign a violation x to every mora with strength y that dominates a consonant such that $x=y-0.5$. weight=10

In addition to the two constraints in (13), we assume the constraint $DEP\mu$ in (14a) which is violated by any addition of strength to a mora, and the constraint $MAX\mu$ in (14b) which is violated by any deletion of strength of a mora. Further, we assume the constraint $*FLOAT$ in (14c) which is gradiently violated by moras which are not integrated into the syllable structure but remain floating. $DEP\mu$ and $MAX\mu$ receive particularly high weight to avoid adding or subtracting activity to a weak or strong mora to satisfy $*V^{\mu 0.5}$ or $*C^{\mu 1}$. $*FLOAT$ has a weight of 7 which is lower than both $*V^{\mu 0.5}$ and $*C^{\mu 1}$. This weighting ensures that weak moraic linking elements remain floating instead of associating with a vowel, and strong moraic linking elements remain floating instead of associating with a consonant.

- (14) a. $\boxed{\text{DEP}\mu}$ Assign a violation x for every mora with strength x in the output that is not present in the input. weight=20
- b. $\boxed{\text{MAX}\mu}$ Assign a violation x for every mora with strength x in the input that is not present in the output. weight=20
- c. $\boxed{*FLOAT}$ Assign a violation x for every element of strength x which is not dominated by a syllable and does not dominate a segment in the output. weight=7

We establish the computation of consonant gemination at the compound boundary using the example in (15). This example shows a subcompound consisting of two nouns of which the second has Dravidian origin. Thus, given the origin of the second stem, a linking element in the morphology that is introduced by the compound structure is expounded as the allomorph realised as a weak floating moraic affix: $/(\mu_{0.5})/ \leftrightarrow \text{LE}$.

- (15) [a:nakkut̪ira]
 a:na-($\mu_{0.5}$)-kut̪ira
 elephant-LE-horse_[DRAVIDIAN]
 ‘horse that is like an elephant’ (subcompound)(p.90)

The tableau for the example (15) is given below in (16). Some notes on the representation of the linguistic elements are in order. In this and all following tableaux the entire compound is shown segmentally. Suprasegmentally, in order not to clutter the tableaux, the moras are only represented in superscript form for the segments that are possible targets for the floating moraic linking elements. Possible targets are segments that are at the compound boundary. A long vowel is represented with two strong moras, i.e $V^{\mu^1\mu^1}$, and a consonant geminate with a weak mora and a dot underneath the consonant, indicating the syllable boundary, i.e $C^{\mu_{0.5}}$. Elsewhere long vowels are represented by the length symbol : and consonant geminates by a doubled consonant.

Tableau (16) shows how the most harmonic candidate in (16a) associates the weak floating mora to a consonant, thereby leading to consonant gemination. This output does not violate any of the assumed constraints. Output candidate (16b) associates the weak mora to a vowel, thereby lengthening the vowel. This candidate incurs a gradient violation of 0.5 to the constraint $*V^{\mu_{0.5}}$ worsening the harmony score considerably to -5. Similarly, in output (16c) the mora associates with a vowel. However, in this output, the mora is strengthened to 1. This output no longer violates $*V^{\mu_{0.5}}$ yet the addition of 0.5 to the activity of the mora incurs a gradient but fatal violation of 0.5 to $\text{DEP}\mu$ and leads to a harmony score of -10. Finally, output (16d) fails to integrate the floating mora, thereby incurring a gradient violation of 0.5 to $*FLOAT$ which causes a harmony score of -3.5.

(16) *Consonant gemination at the compound boundary:*

	a:na ^{μ1} (^{μ0.5})kuṭira	DEP ^μ 20	MAX ^μ 20	*V ^{μ0.5} 10	*C ^{μ1} 10	*FLOAT 7	$\mathcal{H}$
☞	a. a:na ^{μ1} k ^{μ0.5} u.ṭi.ra						0
	b. a:na ^{μ1} μ ^{0.5} .ku.ṭi.ra			0.5			-5
	c. a:na ^{μ1} μ ¹ .ku.ṭi.ra	0.5					-10
	d. a:na ^{μ1} (^{μ0.5}).ku.ṭi.ra				0.5		-3.5

Vowel lengthening at the compound boundary is shown in the example (17). This example also consists of a subcompound with two nouns. This time the second stem has Sanskrit origin. In this example, the allomorphic linking element that is sensitive to Sanskrit stems is chosen which is exponed by a strong floating mora: $/(\mu_1)/ \leftrightarrow \text{LE}$.

- (17) [tara:ka:ntan]
 tara-(^{μ1})-ka:ntan
 Tara-LE-husband_[SANSKRIT]
 ‘Tara’s husband’ (subcompound)(p.36 Mohanan 1982)

The relevant tableau of example (17) is given below in (18). In this configuration, output (18a) which associates the strong mora with a consonant, ends up with a harmony score of -5. This is due to the gradient violation by 0.5 of *C^{μ0.5}. Unlike in the previous tableau which had a weak input floating mora, the strong mora is more harmoniously integrated such that it associates with a vowel. This is exemplified by output (18b) which does not violate any of the proposed constraints because it associates the mora, satisfying *FLOAT, the fully active mora dominates a vowel not a consonant, satisfying *C^{μ0.5} and *V^{μ0.5} and no strength is added to or subtracted from the mora, satisfying DEP^μ and MAX^μ. In the output candidate (18c) the mora dominates a consonant and strength is subtracted from the mora. The subtraction of strength gradiently violates MAX^μ by 0.5 which worsens the harmony score considerably to -10. Output candidate (18d) leaves the mora afloat which fully violates *FLOAT causing a harmony score of -7.

(18) *Vowel lengthening at the compound boundary*

	$\text{ta.ra}^{\mu 1} \textcircled{\mu 1} \text{ka:n.tan}$	$\text{DEP}\mu$ 20	$\text{MAX}\mu$ 20	$*V^{\mu 0.5}$ 10	$*C^{\mu 1}$ 10	$*\text{FLOAT}$ 7	$\mathcal{H}$
a.	$\text{ta.ra}^{\mu 1} \text{ḱ}^{\mu 1} \text{a:n.tan}$				0.5		-5
 b.	$\text{ta.ra}^{\mu 1 \mu 1} \text{.ka:n.tan}$						0
c.	$\text{ta.ra}^{\mu 1} \text{ḱ}^{\mu 0.5} \text{a:n.tan}$		0.5				-10
d.	$\text{ta.ra}^{\mu 1} \textcircled{\mu 1} \text{.ka:n.tan}$				1		-7

The derivations in tableaux (16) and (18) show that the two lengthening processes in Malayalam are derived by a single constraint ranking in the phonology. This is achieved by the assumption that the two lengthening outcomes originate from an allomorphic distinction in the morphology reflected in the phonology by a difference in moraic strength. This distinction leads to different effects in the phonological component, thereby deriving either consonant gemination or vowel lengthening.

The analysis using allomorphic moraic linking elements in combination with the proposed constraint weighting predicts that a strong floating mora in Sanskrit environments will never associate with a consonant and vice-versa that a weak floating mora in Dravidian environments will never end up associating with a vowel. This follows from the weighting of constraints $*V^{\mu 0.5}, *C^{\mu 1} \gg *F\text{LOAT}$. It is more harmonic to leave a mora floating than to associate a weak mora with a vowel or a strong mora with a consonant. This result is borne out. In example (19a), the second noun has Sanskrit origin therefore the moraic linking element is strong. However, there is no available vowel at the compound boundary for the strong mora to associate with and, even though there is a stem-initial voiceless plosive at the boundary, no consonant gemination is observed⁴. Similarly, in (19b) the second noun has Dravidian origin, in this case, the weak moraic linking element is selected. There is no voiceless plosive available for the mora to associate with and glides cannot be geminated in Malayalam. Crucially, there is a stem-final vowel at the boundary, yet it is not lengthened.

- (19) a. [kulirka:ttə]
 kulir-kar:tt
 cold-breeze_[SANSKRIT?]
 ‘cool breeze’ (subcompound) (p.85)

⁴Malayalam phonology readily allows clusters of sonorants followed by geminated plosives in subcompounds, so there is a priori no phonological reason for the non-realisation of the geminate consonant.

- b. [b^harjawɪːtə]
 b^harja-wiːt
 wife-house_[DRAVIDIAN]
 ‘wife’s house’ (subcompound) (p.81)

In both cases in (19) the combination of the choice of allomorph, the low weight of *FLOAT and the high weight of *V^{μ0.5} and *C^{μ1} cause the most harmonic output to be the candidate that surfaces with a floating mora and no phonological effect of lengthening is observed. In section 3.2, we will see that a weak mora may associate with the epenthetic schwa, but only because a higher weighted constraint against illicit consonant clusters in Malayalam outweighs *V^{μ0.5}.

2.1 Weak and strong moras in Malayalam

Up to this point, we have postulated that vowels in Malayalam are dominated by strong moras while consonant geminates are dominated by weak moras. This postulation allowed us to derive the lexically enforced difference between vowel lengthening and consonant gemination at the compound boundary. In this section, we aim to provide independent evidence for the difference in moraic strength. We focus on two issues in Malayalam phonology, stress and minimal word effects.

According to Hayes (1989), a geminated consonant is a consonant that is associated with a mora which supplies weight to a syllable. Thus syllables with consonant geminates are heavy syllables. Syllables with long vowels also constitute heavy syllables. It follows that geminate consonants should pattern with long vowels concerning weight-sensitive processes. However, in Malayalam, this is only partly borne out.

Stress is a weight-sensitive phenomenon in Malayalam. Long vowels attract stress. Take example (20) which shows that primary stress normally falls on the first syllable of the word, see (20a). However, if the second syllable contains a long vowel and the first syllable is short, primary stress falls on the second syllable, see (20b).

- (20) a. [ˈpaɾaːti]
 ‘searched’
 b. [paˈraːti]
 ‘complaint’ (p.112)

Intriguingly, a syllable containing a consonant geminate will not attract stress, as shown in the example (21b). Here, the second syllable contains a consonant geminate and the first syllable is short, yet, stress is realised on the first syllable of the word.

- (21) a. paˈraːti ‘complaint’
 b. ˈpaɾaːtti ‘scattered’ (Mohan 1989: 611)

Despite the consonant geminate assumedly contributing a second mora to the syllable just like a long vowel would, stress is not realised on this heavy syllable. It has been argued because of these data that geminated consonants in Malayalam are not moraic (Broselow et al. 1997). However, geminate consonants do contribute a mora from the perspective of the minimal word condition in Malayalam. Mohanan (1989) and Namboodiripad et al. (2015) show that a word in Malayalam must be at least bimoraic. Thus, a word may consist of a single syllable with a long vowel, cf. (22a) but not a short vowel, cf. (22b). Note that in the examples in (22) the final schwa does not contribute a mora. We follow Mohanan (1989) who assumes that schwa epenthesis applies to resyllabify a coda consonant⁵ after the minimal word condition is evaluated. Finally, consider the example in (22c), which consists of a single syllable with a final geminated consonant. This configuration is a possible word in Malayalam. Thus, for the case of the minimal word condition, geminates in Malayalam must provide a second mora to the syllable.

- (22) a. pa:lə ‘milk’
 b. *palə
 c. pallə ‘tooth’

The evidence from weight-sensitive stress assignment and the minimal word condition conflict with respect to the moraic status of geminated consonants. On the one hand, Malayalam geminates cannot contribute a mora otherwise they would incorrectly attract stress. On the other hand, Malayalam consonant geminates need to be able to contribute a mora for the minimal word condition.

This conflict is resolved under the assumption that moras come in different flavours implemented as different strengths. A vowel is associated with a strong mora that is fully active and a geminate consonant with a weak mora, that has the activity 0.5. Thus, the phonological grammar can distinguish between and be sensitive to different strengths of moras. The minimal word condition must be sensitive to the amount of moras in the word, not their strength, while stress is sensitive to the strength of moras. Stress is only attracted to syllables with the moraic strength of (at least) two, which is exactly two vowels but cannot be a vowel and a geminated consonant⁶.

Summing up the analysis thus far, we have established that the lexically determined difference between vowel lengthening and consonant gemination can be treated as lexically conditioned allomorphy of a morphological linking element that is realised as either a strong or a weak moraic affix. Further, the differentiation between strong moras that are associated with vowels and weak moras that are

⁵Only nasals may be coda consonants in Malayalam (Mohanani 1986)

⁶This makes interesting predictions on the status of epenthetic ə as it is implemented in the framework of GSR. Due to the constraint $\text{DEP}\mu$ outweighing $*V^{\mu 0.5}$ it will always be preferred to epenthesise a mora that is particularly weak. This means that epenthetic material will never be strong enough to attract stress. The data presented in Mohanan (1986) does not provide evidence to test this prediction as an epenthetic schwa is never long. Therefore they can never attract stress.

associated with consonants receives independent support from the general phonology of Malayalam. Thus, we have explained two of the six configurations that lead to different phonological effects at the compound boundary of Malayalam. In the next section, we shift the focus from the morphology to the phonological influence on lengthening at the compound boundary.

3 The influence of phonology on moraic affixation

In this section we tackle three independent problems for the moraic affixation theory that all boil down to a phonological influence on its realisation. First, the structural restriction on the realisation of the linking element as consonant gemination in co-compounds. Second, the bleeding interaction of schwa epenthesis on gemination. And third, the special status of *t̪* and *r* which can override the bleeding interaction.

3.1 Mora affixation is sensitive to prosodic structure

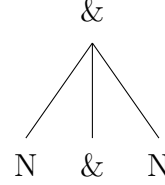
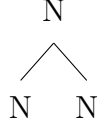
Consonant gemination is restricted to apply within subcompounds, and is blocked in co-compounds. Example (23) repeats the issue. In a subcompound structure presented in example (23a), the initial consonant of the second stem is geminated, as expected. In a co-compound structure presented in example (23b), the initial consonant of the second stems unexpectedly does not geminate.

- (23) a. [petti^{pp}atta:jam^{na}la]
- petti-patta:jam-kal
- box-grainbin_[DRAVID]-PL
- ‘grainbins used as boxes’ (subcompound)(p.90)
- b. [petti^patta:jam^{na}la]
- petti-patta:jam-kal
- box-grainbin_[DRAVID]-PL
- ‘boxes and grainbins’ (co-compound)(p.90)

We claim that consonant gemination is blocked in co-compounds due its to sensitivity to prosodic structure. Prosodic parsing of co-compounds differs from prosodic parsing of subcompounds due to a difference in their morphological structure. In co-compounds, each noun is parsed into separate prosodic words, whereas in subcompounds the entire compound is parsed into a single prosodic word. Thus, in co-compounds, there is an internal prosodic word boundary which is non-existent in subcompounds. We argue that the realisation of geminated consonants is sensitive to this boundary. The proposed morphological structures of the compounds are given in (24). Subcompounds are balanced compounds that consist of two nominal elements, see (24a). Co-compounds have a coordinate structure, they consist of two nominal elements headed by an abstract & element,

see (24b) (see Di Sciullo 2005, 2011 for arguments to assume a functional head in compounds and Nóbrega 2020 to assume an abstract & head in cocompounds).

- (24) a. *Morphological structure of a subcompound:* b. *Morphological structure of a co-compound:*



The two morphological structures parse distinct prosodic structures. A subcompound constitutes a single prosodic word, see (25a), while in a co-compound each nominal element constitutes a single prosodic word, see (25b). These domains also match the stress domain in Malayalam compounds. We refer to Author & Author (2024) for a full analysis of both prosodic parses in match theory (Selkirk 2011).

- (25) a. *Parsing of a subcompound:* b. *Parsing of a co-compound:*
 $(_{\omega} \text{ N N})$ $(_{\omega} \text{ N }) (_{\omega} \text{ N })$

The prosodic parses coincide with the stress domains of Malayalam compounds. Subcompounds constitute a single domain for stress while each head in the co-compound constitutes a single domain for stress, see Mohanan (1986) and Author & Author (2024) for more details.

Given the prosodic structures in (25), a geminated consonant cannot be realised at the left edge of a prosodic word boundary, which derives the blocking of consonant gemination in co-compounds. Co-compounds exhibit an additional prosodic boundary between the two nominal elements which brings about a compound-internal left-edge prosodic word boundary. The blocking of consonant gemination is implemented into the theory by the constraint $\boxed{*(_{\omega} \text{ C}^{\mu})}$ in (26).

- (26) $\boxed{*(_{\omega} \text{ C}^{\mu})}$: Assign a violation x to every consonant that is at the left edge of a prosodic word ω and dominated by a mora of strength x .
weight=20

The non-realisation of the moraic linking element in co-compounds is derived in tableau (27). The linking element is realised by a weak mora due to the Dravidian origin of the second stem. This weak mora can be integrated into the syllable structure by associating with a vowel in candidate (27d). This integration avoids creating a consonant geminate, however, it gradiently violates $*V^{\mu 0.5}$ leading to a harmony score of -5. Candidate (27b) associates the mora with a consonant thereby creating a geminate. Crucially, in this candidate, the edge of a prosodic word boundary intervenes and therefore creates a configuration where a geminated consonant is realised at the left edge of a prosodic word boundary. This configuration gradiently violates high weighted $*(_{\omega} \text{ C}^{\mu})$ thereby causing a fatal harmony

score of -10. The prosodic word boundary in the compound need not be parsed as shown by candidate (27c). This candidate avoids a compound-internal left edge boundary by parsing an asymmetric recursive prosodic word. However, this parsing does not match a nominal element with a prosodic word thus violating the match constraint $\text{MATCH}(\text{N}, \omega)$ leading to a score of -4. Candidate (27a) ends up most harmonic. This candidate parses all nominal elements matching them with a prosodic word but does not integrate the moraic affix into the syllable structure. As a consequence, the constraint $\ast\text{FLOAT}$ is gradiently violated which leads to the most harmonic score of -3.5.

(27) *Derivation of the gemination blocking effect in co-compounds*

[& [N petti] ($\mu_{0.5}$) [N patta:jam]]		$\ast(\omega C^\mu$ 20	$\ast V^{\mu 0.5}$ 10	$\ast\text{FLOAT}$ 7	$\text{MATCH}(\text{N}, \omega)$ 4	$\mathcal{H}$
	a. (ω petti) ($\mu_{0.5}$) (ω patta:jam)			0.5		-3.5
	b. (ω petti) (ω ppatta:jam)	0.5				-10
	c. (ω (ω petti) ppatta:jam)				1	-4
	d. (ω petti:) (ω patta:jam)		0.5			-5

The non-realisation of the weak moraic affix in co-compounds is a reflex of the morphological structure of co-compounds, visible to the phonology by virtue of indirectly matched prosodic structure. The constraint $\ast(\omega C^\mu$ ensures that no geminated consonant is realised at a left edge of a prosodic word boundary. This constraint extends to the phonology outside of compounding. It ensures that no word-initial geminates surface which holds for the entire phonology of Malayalam (Asher & Kumari 1997). Although there are several derivational prefixes that introduce derived geminates in root-initial position (Mohanani 1986), see example (28), there are no word-initial consonant geminates. In example (28), the derivational prefix $/\text{nir}^{(\mu 0.5)}/$, *without*, induces gemination of the root-initial velar plosive $/g/$. We assume, analogous to the weak moraic linking element, that this prefix is underlyingly specified with a weak floating mora.

- (28) [nirgguṇam]
 $\text{nir}^{(\mu 0.5)}$ -guṇam
 without-quality
 ‘without quality’ (p.93)

The constraint $\ast(\omega C^\mu$ could conflict with the datum in (28). However, once derivational prefixes are parsed in the same prosodic word as the root, $\ast(\omega C^\mu$ is satisfied. The stress domain of Malayalam corroborates this parse. Derivational prefixes are inside of the stress domain of Malayalam. Although Mohanani (1986) does not explicitly state this, his stratal model derives it and his examples allude to derivational affixes being inside the stress domain (cmp. [‘su.k^ham] *well being* vs. [‘a.su.k^ham] *illness* with a stressed $/a/$ derivational prefix).

3.2 Mora affixation to schwa

Schwa epenthesis at the compound boundary bleeds consonant gemination. Mohanan (1986) shows that a schwa is epenthesised between two consonants at a compound boundary when they would form an illicit consonant cluster. This is exemplified by the compound in (29). Here, two plosives become adjacent due to compounding. The cluster is resolved by epenthesising a schwa.

- (29) [camp $\boxed{\text{ə}}$ p a:tɾam]
 camp-($\mu_{0.5}$)-pa:tɾam
 copper-LE-vessel_[DRAVIDIAN]
 ‘copper vessel’(subcompound)(p.85)

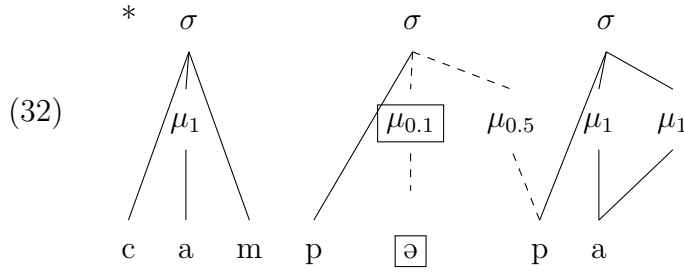
The second stem in (29) has Dravidian origin, thus the linking element is expected to surface as a geminated consonant. However, the initial consonant of the second stem /pa:tɾam/ remains a singleton consonant in (29). Thus, schwa epenthesis bleeds stem-initial consonant gemination. Example (30) shows that the initial consonant in /pa:tɾam/ does surface as a geminate when no schwa is epenthesised.

- (30) [pa:l $\boxed{\text{pp}}$ a:tɾam]
 pa:l-($\mu_{0.5}$)-pa:tɾam
 milk-LE-vessel_[DRAVIDIAN]
 ‘vessel for milk’(subcompound)(p.85)

The bleeding interaction between epenthesis and gemination follows straightforwardly from the assumption that a moraic affix realises a linking element at the compound boundary. When the segment schwa is epenthesised to break up an illicit consonant cluster the moraic linking element associates with [ə] instead of with the stem-initial consonant. This is sketched in (31). In this and all future examples epenthetic material is boxed and epenthetic association lines are dashed.

- (31)
-

Alternatively, the moraic linking element is associated with the stem-initial consonant. In this case, the epenthesised [ə] that breaks up the illicit cluster, is provided with an epenthetic (weak) mora so that it can be realised. This is shown in (32).



Given the constraints and their weighting in addition to the constraint *P.P in (33) which is penalised by adjacent plosive consonants, the grammar correctly derives (31) and excludes (32).

- (33) *P.P: Assign a violation x for every plosive of strength y that is syllable adjacent to a plosive of strength z such that $x=(y+z)/2$. weight=100

Crucial to the constraint interactions is the high weight of $\text{DEP}\mu$ and gradient violation of $V^{\mu 0.5}$. Tableau (34) shows the computation. The most harmonic output (34a) associates the weak mora with the epenthetic [ə] and the consonant is realised as a singleton consonant. Associating the floating mora with a vowel violates $*V^{\mu 0.5}$ gradiently by 0.5. This leads to a harmony score of -5. Output (34b) differs from (34a) in having epenthesised the schwa but nonetheless geminated the consonant. This results in an output form where the weak floating mora dominates a consonant thereby satisfying the constraint $*C^{\mu 1}$ that is not shown in this tableau. However, as a consequence, the epenthetic vowel must be dominated by an epenthetic mora. In this case, the mora is particularly weak, it thus violates $\text{DEP}\mu$ to a low degree by 0.1, but at the same time violates $*V^{\mu 0.5}$ to a high degree by 0.9. Both violations lead to a worsened harmony score of -11⁷. The third output (34c) refrains from epenthesising a vowel to break up the consonant cluster and consequently incurs a full violation of *P.P and a particularly bad harmony score of -100. Thus, the most harmonic output candidate in (34a) derives the bleeding interaction.

⁷Other candidates that vary in how much strength is added to the epenthesised mora are not considered in the tableau because they will never be more harmonious than output (34a) due to the high weight of $\text{DEP}\mu$

(34) *Interaction of ə-epenthesis and compound gemination:*

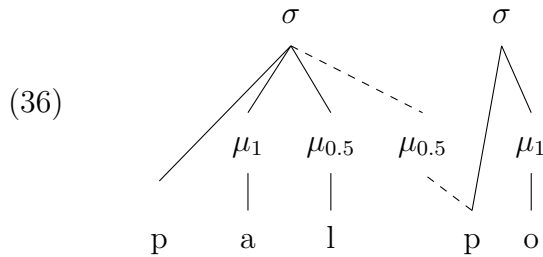
camp ^(μ0.5) patram	*P.P 100	DEPμ 20	*V ^{μ0.5} 10	*FLOAT 7	$\mathcal{H}$
☞ a. camp ^[ə] ^{μ0.5} .patram			0.5		-5
b. camp ^[ə] ^[μ0.1] p ^{μ0.5} atram		0.1	0.9		-11
c. camp ^(μ0.5) .patram	1			0.5	-103.5

Tableau (34) exemplifies that the constraint $*V^{\mu 0.5}$ can be violated in Malayalam but only to satisfy a higher weighted constraint in return. If a schwa has to be epenthesised to break up an illicit consonant cluster, the weak moraic affix associates with the vowel. It is more costly to associate with the consonant because of the high weight of $DEP\mu$, and it is more costly to remain floating due to the high weight of $*P.P$.

There is a second interaction with stem-initial consonant gemination which similarly follows straightforwardly from an analysis employing a moraic affix. When the first stem ends in an underlying geminate consonant and the second stem has Dravidian origin and begins with a voiceless plosive consonant, the underlying consonant geminate is realised as a singleton consonant whereas the initial consonant of the second stem geminates, see (35). Mohanan (1986) calls this phenomenon sonorant degemination. Speaking in terms of rule interaction, we can observe that consonant gemination feeds sonorant degemination.

- (35) [palppoti]
 pall-(^{μ0.5})-poti
 tooth-LE-powder_[DRAVIDIAN]
 ‘tooth powder’ (subcompound)(p.87)

The second stem in (35) has Dravidian origin, therefore a weak moraic affix realises the linking element. The moraic affix associates with the stem-initial consonant but doing so is only possible if the underlying geminate consonant loses its geminate status. Example (36) demonstrates this. Crucially, there is no way for the underlying consonant to be realised as a geminate simultaneously with the derived geminate without violating the No Crossing Constraint (Goldsmith 1976).



The only way to realise the stem-final consonant as a geminate is if a syllable nucleus is epenthesised to accommodate the underlying geminate⁸. The sketched example in (37) shows this configuration⁹. A schwa is epenthesised to realise the stem-final consonants as a geminate and the moraic linking element associates with the epenthetic schwa. This potential output shows the expected bleeding interaction between schwa epenthesis and stem-initial consonant gemination.

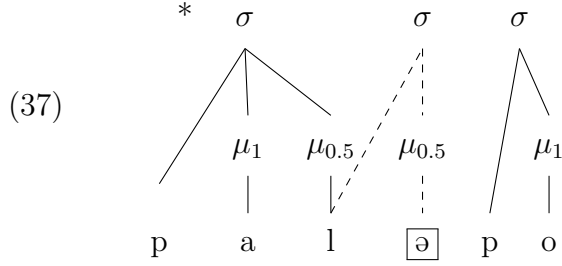


Tableau (38) shows how the output form in (36) in candidate (38b) is computed as most harmonic. This candidate violates none of the assumed constraints. The weak mora is associated with a consonant which satisfies both $*V^{\mu 0.5}$ and $*C^{\mu 1}$ and the coda consonant cluster created by gemination of the stem-initial plosive is harmonic in Malayalam¹⁰. Output (37) which is represented by candidate (38a) incurs a gradient but fatal violation of $*V^{\mu 0.5}$ because a vocalic element is dominated by a mora that is not fully activated. Output (38c) leaves the mora floating which gradiently but fatally violates $*FLOAT$.

⁸Another alternative would be realising the underlying geminate by simply adding an association line from segment to the following syllable: [pal.lpoṭi]. However, this creates a complex onset which is never observed in Malayalam. I therefore do not consider this candidate further.

⁹Interestingly, Mohanan (1986) presents an example of a compound with the same nouns as in (35) but with a different meaning. The example surfaces with an epenthesised schwa which bleeds stem-initial gemination: pall-poṭi tooth-powder → [palləpoṭi] ‘powdered teeth’. It is not clear to us how this difference comes about and this is the only example where a difference in the phonological form reflects a difference in the meaning of a compound. Further research is needed to determine the details of this observation.

¹⁰ $*P.P$ is not violated by this candidate because there is only one plosive which constitutes the coda and onset of two separate syllables, straddling the syllable boundary. $*P.P$ is only violated when two plosives become adjacent at a syllable boundary.

(38) *Interaction of underlying consonant geminates and compound gemination*

	$\text{pal}^{\mu 0.5}(\mu 0.5)\text{poti}$	$*V^{\mu 0.5}$ 10	$*\text{FLOAT}$ 7	$\mathcal{H}$
a.	$\text{pal}^{\mu 0.5}[\text{ə}]^{\mu 0.5}.\text{poti}$	0.5		-5
b.	$\text{pal}^{\mu 0.5}\text{p}^{\mu 0.5}\text{oti}$			0
c.	$\text{pal}^{\mu 0.5}(\mu 0.5).\text{poti}$		0.5	-3.5

Under the assumption of moraic affixation, bleeding and feeding interactions with stem-initial gemination are derived. The crucial insight is that the weak floating mora is flexible in its association. It can depart from its most harmonic association with a consonant if this avoids violations of higher-weighted syllable well-formedness constraints, such as $*P.P$. If no higher-weighted constraints conflict, the mora will only associate with a consonant and not with a vowel.

3.3 Moraic affixation to t and r

Up to this point, we have shown how moraic affixation in the form of a lexically conditioned allomorphic linking element derives four configurations: vowel lengthening via a lexically conditioned strong allomorphic variant of the linking element, consonant gemination via a lexically conditioned weak allomorphic variant of the linking element, blocking of gemination in co-compounds due to a prosodic restriction on geminates, and finally, schwa epenthesis bleeding gemination via a weak moraic affix which associates with an epenthetic schwa to avoid an illicit consonant cluster. Two configurations remain, the first concerns unexpected stem-final gemination of $/t/$ and $/r/$ and the second schwa epenthesis which does not bleed gemination of final $/t/$ and $/r/$. The examples in (39) show that in all cases where $/t/$ or $/r/$ are stem-final at a compound boundary and the second stem has Dravidian origin, $/t/$ and $/r/$ surface as geminated consonants. Example (39b) shows that even if the initial consonant of the second stem is a valid target for gemination, it is the final consonant $/t/$ that is geminated. Crucially, this example also makes clear that only one instance of gemination surfaces which further supports the analysis of lengthening arising through a single moraic linking element. It is not the case that in (39b) both the final $/t/$ and the initial plosive are geminated, which would be expected if all lengthening is caused by phonological rules¹¹.

¹¹In Mohanan (1982) the two rules of gemination were indeed collapsed into a single mirror-image rule in a disjunctive relation. Later in Mohanan (1986) this relation is revised and the non-applicability of stem-initial gemination in (39) is attributed to the phonological environment. Stem-initial gemination only applies if the segment preceding the initial obstruent of the second noun is a nonnasal sonorant. Assuming that schwa epenthesis applies after stem-initial gemination the non-applicability of stem-initial gemination in (39) is accounted for.

- (39) a. [ka: tt a:na]
 ka:t-($\mu_{0.5}$)-a:na
 forest-LE-elephant
 ‘untamed elephant’
- b. [a: tt ə t ozuttə]
 a:t-($\mu_{0.5}$)-tozuttə
 sheep-LE-stable
 ‘sheep shed’
- c. [cawa tt əkot̪ta]
 cawar-($\mu_{0.5}$)-kot̪ta
 trash-LE-basket
 ‘trash can’ (subcomp)(p.91)

Usually, the final consonant of the first stem in a compound is not a target for gemination. Even if phonologically salient, a stem-final consonant at the compound boundary will not geminate. Example (40) shows a stem-final plosive /k/ at the compound boundary. The second stem in the compound has Dravidian origin, thus we expect gemination to occur, however, the plosive surfaces as a singleton consonant despite the phonological environment allowing a geminated consonant in this position.

- (40) [wira k atuppə] *[wira kk atuppə]
 wirak-($\mu_{0.5}$)-atupp
 firewood-LE-stove_[DRAVIDIAN]
 ‘stove that uses firewood’

To arrive at the observed positional restriction, we propose an identity constraint that is penalised if an underlying singleton consonant at the right edge of a morphological boundary surface as geminated consonants, see (41). The constraint is weighted comparably low, as it can be violated by stem-final /t/ and /r/ gemination.

- (41) IO-IDC: Assign a violation x for every undominated consonant with strength x at the right edge of a morphological boundary] in the input, that is dominated by a mora in the output.
 weight = 4

The following tableau in (42) derives the example in (40). Candidate (42a) which associates the mora with the stem-final consonant fully violates the constraint IO-IDC] because an underlying singleton consonant at the right edge of a morphological boundary is dominated by a mora in the output. This violation leads to an overall harmony score of -4. The second candidate in (42b) does not improve on this harmony score. This candidate associates the moraic linking element to a vowel, thereby gradiently violating $*V^{\mu_{0.5}}$ by 0.5 which leads to an

overall harmony score of -5. The final and most harmonic candidate (42c) refrains from phonologically integrating the mora. This leads to a gradient violation of *FLOAT, yet to the overall most harmonic score of -3.5.

(42) *No stem-final gemination at the compound boundary:*

[a:wirak] ^(μ0.5) [a ^{μ1} tupp]	DEP μ 20	*V μ 0.5 10	*FLOAT 7	IO-IDC] 4	$\mathcal{H}$
a. wirak ^{μ0.5} a ^{μ1} tupp				1	-4
b. wira.ka ^{μ1μ0.5} tupp		0.5			-5
 c. wira.k ^(μ0.5) a ^{μ1} tupp			0.5		-3.5

Note that in the tableau above, the underlying stem-final geminate of the second stem in the compound does not violate the constraint IO-IDC] because it is dominated by a mora in the input. The definition of IO-IDC] ensures that only derived geminates at the right edge of a morphological boundary are prohibited from surfacing¹².

Returning to the consonants /t/ and /r/, these do surface as geminates if they are in stem-final position at the compound boundary, as established by examples (39). A further issue raised by the example in (39b), repeated below in (43) for convenience, is the fact that gemination is not bled by schwa epenthesis.

(43) [a:tt̪ətozutt̪ə]
a:t̪-(μ 0.5)-tozutt̪
sheep-LE-stable
‘sheep shed’

¹²One note about derived geminates outside of compounding is in order. There are certain morphological environments which trigger the gemination of a final stem consonant, like the CAUS and TR suffixes. The causative for example is realised as follows: /muruk-(μ 0.5)um/ tighten-CAUS → [murukkum]. According to the constraint IO-IDC], this outcome is unexpected given that the geminates are in stem-final position at the right edge of a morphological boundary. To derive the restriction in compounds yet application of stem-final gemination in inflectional suffixes, we follow Mohanan (1986) in assuming that inflectional affixes are in a different stratum to compounding. Two additional arguments point towards this stratal distinction: The first argument is morphological. Inflectional affixes always occur outside of compounds, never inside of compounds. Thus, no inflectional affixes can be added before compounding has applied. The second argument is phonological. Inflectional affixes do not contribute to the stress domain of Malayalam. Unlike derivational affixes, inflectional affixes are irrelevant to stress placement and are always unstressed, see Mohanan (1986) for more details on the stress domains of Malayalam. Both facts follow if inflectional affixes are added at a later stratum than the stratum where compounding takes place. Then, it also follows that inflectional affixes undergo a different phonology, where the constraints are weighted such that IO-IDC] is demoted to allow stem-final gemination.

The current constraint weighting predicts for an underlying form /a:t-($\mu_{0.5}$)-tozutt/ that schwa epenthesis bleeds any gemination, resulting in the unattested output form *[a:t($\mu_{0.5}$)tozuttə]. To avoid such an output and permit stem-final gemination for /t/ and /r/, we introduce two additional constraints that single out the consonants /t/ and /r/ and enforce their gemination, see (44)¹³. These constraints enforce a particular non-faithful realisation of the consonants /t/ and /r/.

- (44) a. $\boxed{t^{\mu_{0.5}}}$ Assign a violation x for every t with strength x in the input that is not dominated by a mora of strength 0.5 in the output. weight=11
- b. $\boxed{r^{\mu_{0.5}}}$ Assign a violation x for every r with strength x that is not dominated by a mora of strength 0.5 in the output. weight=11

Both constraints outweigh IO-IDC] considerably and $*V^{\mu_{0.5}}$ minorly. This weighting enforces that the moraic linking element associates with /t/ and /r/ at the right edge of a morphological boundary, instead of remaining afloat or associating with an epenthetic schwa.

Tableau (45) shows the computation of example (39a), where there is no need to epenthesise a schwa because there is no illicit consonant cluster created at the compound boundary, *P.P is vacuously satisfied. Two candidates are considered, candidate (45a) associates the mora which creates a t geminate. This geminate violates the constraint IO-IDC], yet still has the most harmonic score of -4. The competing candidate (45b) does not integrate the mora into the syllable structure. In a configuration without /t/, this candidate would become most harmonic. However, the existence of an ungeminated stem-final /t/ violates the constraint $t^{\mu_{0.5}}$. This violation and the violation of *FLOAT generate a considerably worse harmony score of -14.5.

¹³The definition of the constraints $t^{\mu_{0.5}}$ and $r^{\mu_{0.5}}$ ensure that t and r only geminate when there is an underlying floating mora introduced in the morphology. This is because the constraint is satisfied only when a mora with strength 0.5 is associated with either t or r. If there is no floating mora in the input, the constraint can only be satisfied by epenthesising 0.5 moraic strength, which incurs fatal violations of high-weighted $DEP\mu$. The tableau below shows the computation of a stem with final /t/ outside of a compounding context. The most harmonic output has not geminated the final /t/ which incurs a violation of $t^{\mu_{0.5}}$. The competing output candidate b. geminates the final t by epenthesising a mora with 0.5 strength, which avoids the violation of $t^{\mu_{0.5}}$. However, epenthesis of moraic strength gradiently violates $DEP\mu$ which, adding the violation of ID-IDC] for the stem-final geminate, leads to a higher harmony score of -14. Violations of $DEP\mu$ can be avoided by epenthesising less moraic strength, as candidate c. does. But crucially, epenthesis of less than 0.5 does not satisfy the constraint $t^{\mu_{0.5}}$. Thus, candidate c. also violates $t^{\mu_{0.5}}$ and $DEP\mu$ as well as ID-IDC].

	[a:t]	$DEP\mu$	$t^{\mu_{0.5}}$	$*V^{\mu_{0.5}}$	IO-IDC]	$\mathcal{H}$
		20	11	10	4	
J	a. a:t		1			-11
	b. a:t($\mu_{0.5}$)	0.5			1	-14
	c. a:t($\mu_{0.1}$)	0.1	1		1	-17

(45) *Derivation of stem-final gemination*

	$[\text{ka:t}]^{(\mu_{0.5})}[\text{a:na}]$	*P.P 100	$\text{t}^{\mu_{0.5}}$ 11	*FLOAT 7	IO-IDC] 4	$\mathcal{H}$
☞ a.	$\text{ka:t}^{\mu_{0.5}}\text{a:na}$				1	-4
b.	$\text{ka:t}^{(\mu_{0.5})}\text{a:na}$		1	0.5		-14.5

Let us now turn to the example where gemination of a stem-final /t/ and schwa epenthesis co-occur, seen in example (43). As was the case for the previous tableau, the constraint $\text{t}^{\mu_{0.5}}$ ensures that /t/ surfaces as a geminated consonant despite its stem-final position. Additionally, the high weight of *P.P in combination with the weight of $\text{t}^{\mu_{0.5}}$ enforces epenthesis of schwa in addition to gemination. This outcome is represented by output candidate (46b) in tableau (46) below. Despite fully violating IO-IDC], due to the stem-final geminate and gradiently violating $\text{DEP}\mu$ as well as $*V^{\mu_{0.5}}$, due to the epenthesised moraic strength associated with the epenthetic schwa, this candidate ends up most harmonic with a score of -15. Both candidates (46a) and (46d) fail to resolve the illicit consonant cluster, either by associating the mora to the stem-final /t/ or leaving it afloat, without epenthesising a schwa, which causes a fatal violation of *P.P. Candidate (46c) epenthesises a schwa and ensures its realisation by associating the moraic linking element with it. This candidate is most harmonic when no /t/ is involved, thereby giving rise to bleeding of stem-initial gemination. However, the non-geminated /t/ incurs a fatal violation of the constraint $\text{t}^{\mu_{0.5}}$. In combination with the gradient violation of $*V^{\mu_{0.5}}$ incurred by the weak moraic affix associated with schwa, a harmony score of -16 is generated. Candidate (46e) which associates the mora to the stem-initial /t/ of the second stem avoids the violation of IO-IDC] but otherwise concedes a considerable amount of violations by epenthesising a weak mora to associate with the epenthetic schwa and by not geminating stem-final /t/ which leads to the higher harmony score of -22. Thus output candidate (46b) is most harmonic because it avoids a violations of the highly weighted constraint $\text{t}^{\mu_{0.5}}$ by geminating /t/, and *P.P, by epenthesising schwa and thereby concedes only gradient violations of $\text{DEP}\mu$ and $*V^{\mu_{0.5}}$.

(46) *Derivation of stem-final gemination and schwa epenthesis:*

	$[a:t]^{(\mu 0.5)}[tozutt]$	*P.P 100	DEP/ μ 20	$t^{\mu 0.5}$ 11	* $V^{\mu 0.5}$ 10	*FLOAT 7	IO-IDC] 4	$\mathcal{H}$
a.	$a:t^{\mu 0.5}.to.zutt$	1					1	-104
 b.	$a:t^{\mu 0.5}[\emptyset^{\mu 0.1}].to.zutt$		0.1		0.9		1	-15
c.	$a:t[\emptyset^{\mu 0.5}].to.zutt$			1	0.5			-16
d.	$a:t^{(\mu 0.5)}.to.zutt$	1		1		0.5		-114.5
e.	$a:t[\emptyset^{\mu 0.1}].t^{\mu 0.5}o.zutt$		0.1	1	0.9			-22

4 Discussion and conclusion

In the preceding sections, we have shown that all lengthening effects and their interaction with insertion at a compound boundary in Malayalam are derived by a linking element realised as a moraic affix. All six configurations are repeated in the table below. Included in the table are the phonological environment, as well as the lexical and structural environment that cause a different realisation of the moraic affix.

		1	2	3	4	5	6
environment	phonological	V-P/S-P	V-C	any	P-P	t -V/r-V	t -P/r-P
	lexical	[DRAVID]	[SANSKRIT]	[DRAVID]	any	[DRAVID]	[DRAVID]
	= allomorphy	$(\mu 0.5)$	$(\mu 1)$	$(\mu 0.5)$	$(\mu 1)/(\mu 0.5)$	$(\mu 0.5)$	$(\mu 0.5)$
	structural = prosody	subcomp (ω)	any (ω)(ω)/(ω)	co-comp (ω)(ω)	any (ω)(ω)/(ω)	subcomp (ω)	subcomp (ω)
phon. effect	initial gemination	VP:/SP:	–	–	–	–	–
	vowel lengthening	–	V:C	–	–	–	–
	ə-epenthesis	–	–	–	PəP	–	t:əP/t:əP
	final t/r gemination	–	–	–	–	t:V/t:V	t:əP/t:əP

For columns 1 and 2, we have shown that the lexical environment is reflected as lexically conditioned allomorphy of the linking element. A weak moraic allomorph realises the linking element in Dravidian stems which causes consonant gemination in the phonological component due to the constraint $*V^{\mu 0.5}$, and a strong moraic allomorph realises the linking element in Sanskrit stems which causes vowel lengthening in the phonological component due to the constraint $*C^{\mu 1}$. The morphosyntactic, structural environment, which influences the realisation of geminates in the

configuration shown in column 3, is reflected as prosodic structure. We assume that geminates are sensitive to prosodic structure, they cannot occur at the left edge of a prosodic word boundary. In co-compounds a prosodic word boundary is parsed at the compound boundary which causes the non-realisation of a geminate. Finally, the phonological environment determines if a moraic linking element associates with an epenthetic schwa. Column 4 represents the configuration where two plosive consonants (that are not t or r) become adjacent at a compound boundary and a schwa is epenthesised to break up the illicit consonant cluster. In this case, the grammar forces the moraic linking element to associate with the epenthetic schwa. The constraint $*V^{\mu 0.5}$, although gradiently violated by the most harmonic output, in combination with $DEP\mu$ blocks epenthesis of moraic strength from realising the epenthetic schwa. The fact that schwa epenthesis bleeds gemination is not upheld if a stem-final t or r is geminated. In this case, high weighted $t^{\mu 0.5}$ and $r^{\mu 0.5}$ force gemination of $/t/$ and $/r/$ respectively, represented by column 5 and, if phonologically necessary, a schwa will epenthesise and be realised with epenthetic moraic strength, as is the outcome in column 6.

The main argument in our analysis is the unification of all lengthening and insertion effects at the compound boundary. All compound-specific phonology comes about through the realisation of a moraic linking element. Allomorphy, prosody and general phonology influence the phonological realisation of moraic affixation of the linking element. Crucially, the phonology neither needs to refer directly to lexical information nor is it necessary to introduce distinct compound strata, which in Malayalam inevitably leads to the introduction of a stratal loop. Lexical information, distinct strata and the loop are integral parts of Mohanan's (1986) analysis of the phonology and morphology of Malayalam. As established in section 1.1, Mohanan (1986) builds lexical information into his phonological rules to differentiate the application of stem-initial consonant gemination from final vowel lengthening. Compare this to the analysis proposed in this paper, where the difference in consonant gemination and vowel lengthening is reflected in the phonology by representational differences but ultimately determined at a prior stage by the morphology in the form of allomorphy of the linking element. Phonology does require access to morphological bracketing to prohibit gemination of stem-final consonant gemination, but it does not require lexical information. This maintains a more modular approach to the morphophonology interface.

Mohanan (1986) also requires distinct strata to derive the non-realisation of consonant gemination in co-compounds. Mohanan (1986) shows that there are additional phonological effects that support a stratal distinction between the compound types. However, the evidence is quite marginal and prosodic structure suffices to account for the phonological differences. Consider table 2 which represents the main phonological effects at a compound boundary. This table indicates that co-compounds do not undergo consonant gemination, as analysed in section 2, they do not undergo stress assignment as a whole and they are sensitive to different schwa-epenthesis rules.

	stratum 1 (derivation)	stratum 2 subcompounding	stratum 3 co-compounding	stratum 4 (inflection)	postlexical
consonant gemination		✓	–	✓	
vowel lengthening	–	✓	✓		
ə-epenthesis	(✓)	(✓)	✓	✓	
stress assignment	✓	✓	(✓)	–	

Table 2: Summary of all processes relevant for the stratal distinction of compounds

We have established that the different effects of compound gemination are derived by reference to prosodic structure. Further, the differences in stress and schwa epenthesis also follow from the proposed prosodic structures of the compounds. The stress domains in Malayalam compounds differ depending on the compound type. Subcompounds are treated as a single domain for stress, whereas in co-compounds each noun is treated as its own domain for stress. This difference is shown in example (47). The subcompound has a single primary stress on the first syllable of the compound, see (47a), whereas the co-compound has two primary stresses, one each for each noun in the compound, see (47b)¹⁴.

(47) *Stress domain differences in subcompounds versus co-compounds:*

- a. [(pet̪t̪ip̪pat̪, t̪aːjaŋ)ŋalə]
pet̪ti- pat̪t̪aːjam -kal
box- grainbin -PL
'grainbins used as boxes' (subcompound)(p.90)
- b. [(pet̪, ti)(pat̪, t̪aːjaŋ)ŋalə]
pet̪ti- pat̪t̪aːjam -kal
box- grainbin -PL
'boxes and grainbins' (co-compound)(p.90)

These domains of stress follow directly from the proposed prosodic structures introduced in section 2. Subcompounds constitute a single prosodic word domain; (ω N N), therefore they are treated as a single domain for stress. Co-compounds introduce a prosodic word domain for each noun in the compound; (ω N) (ω N), leading to multiple domains for stress¹⁵.

Similarly, the differences in schwa epenthesis also follow from the different prosodic parses. We have seen that a schwa is epenthesised to break up an illicit consonant cluster in subcompounds in section 3.2. This also applies to co-compounds. However, in co-compounds schwa epenthesis seems to overapply.

¹⁴In the examples secondary stress is also indicated. Secondary stress according to Mohanan 1986 falls on all heavy syllables after the first two syllables. Further H pitch is realised on the final syllable of the (prosodic) word, which we also indicate as secondary stress in the examples.

¹⁵Mohanan (1986) also shows that his stratal architecture straightforwardly derives stress in compounds that are embedded in compounds, as well as two opaque interactions between stress and segmental phonological rules. We refer to the analysis in Author & Author (2024), which shows that in a minimal stratal architecture consisting of stem- and word-level morphophonology, the same prosodic parses used here account for the stress domains in complex compounds and for the opaque effects.

Take example (48), where a schwa is epenthesised between a retroflex nasal consonant and a plosive and between a retroflex nasal consonant and a glide. In subcompounds this is an acceptable consonant cluster and not repaired by epenthesis, however in co-compounds schwas are epenthesised.

- (48) [a:ŋ[ə]peŋŋ[ə]wjaɽja:sam]
a:ŋ-peŋŋ-wjaɽja:sam
man-woman-difference
‘difference between men and women’ ((co-comp)subcomp) (p.85)

The overapplication of schwa epenthesis is explained once prosodic structure is taken into account. In co-compounds, each noun in the compound constitutes its own prosodic word domain. Therefore, we expect each noun in a co-compound to behave as it would in isolation. This is evidenced by final epenthesis of schwa. (Prosodic-)Word-finally only /m/ or /n/ are licit coda consonants¹⁶. Elsewhere a schwa is epenthesised to provide a nucleus. Thus, in isolation the words *man* and *woman* are pronounced with a final schwa as [a:ŋə] and [peŋŋə], exactly as they are pronounced in the co-compound. Schwa epenthesis in (48) does not arise to break up a consonant cluster. Instead, it arises to provide a nucleus for an illicit coda consonant.

The phonological differences between sub- and co-compounds thus all follow from the difference in prosodic structure. Thus, there is no need to postulate distinct compound strata and no need for a stratal loop. Obviating the need for a loop is a considerable simplification of stratal architecture. The introduction of the loop constitutes a departure from strict level ordering (Kiparsky 1982, Mohanan 1986) and therefore a considerable weakening of the predictive power of Lexical Morphology and Phonology and its modern counterpart Stratal OT, see also Sproat (1985) who addresses this issue. In this article, we have shown that compound phonology in Malayalam does not provide a strong argument to posit a loop or any kind of less restrictive interleaving of morphology and phonology. We have shown this by addressing the complexities of compound-specific lengthening and insertion in Malayalam. Instead of positing multiple rules and strata to account for the data, we posit that all lengthening and its interactions with insertion at the compound boundary originate from a linking element that is introduced in the morphological component and is realised as a floating mora in the phonological component. The distinct surface properties of the moraic affix follow from three assumptions: First, the linking element has two allomorphs which are both realised by a floating mora but vary in their underlying strength, modeled as gradient symbolic representations. Second, consonant geminates are sensitive to prosodic structure, which differs between co-compounds and subcompounds due to their structural difference. Third, general phonological well-formedness interacts with how the floating mora is realised. Thus, we derive all configurations of lengthening and insertion at the compound boundary with one instance of a linking element.

¹⁶This is the case for casual speech, not fast speech as Mohanan points out. We limit ourselves to casual speech in this analysis, following Mohanan (1986)

References

- Aronoff, Mark & Nanna Fuhrhop. 2002. Restricting suffix combinations in German and English: Closing suffixes and the monosuffix constraint. *Natural Language & Linguistic Theory* 20(3). 451–490.
- Asher, R. E. & T. C. Kumari. 1997. *Malayalam*. London, UK: Routledge.
- Author, A. & B. Author. 2024. [Masked for anonymous review]. *submitted*.
- Broselow, Ellen, Su-I Chen & Marie Huffman. 1997. Syllable weight: convergence of phonology and phonetics. *Phonology* 14(1). 47–82.
- Christdas, Prathima. 1986. On constraining the power of lexical phonology: Evidence from Tamil. In *North east linguistics society*, vol. 17, 9.
- Davis, Stuart & Isao Ueda. 2002. Mora augmentation processes in Japanese. *Journal of Japanese Linguistics* 18(1). 1–24.
- Di Sciullo, Anna Maria. 2005. Decomposing compounds. *Skase Journal of theoretical linguistics* 2(3). 14–33.
- Di Sciullo, Anna Maria. 2011. Why are compounds a part of human language? A view from asymmetry theory. In Rochelle Lieber & Pavol tekauer (eds.), *The Oxford Handbook of Compounding*, 145–177. Oxford: Oxford University Press.
- Fuhrhop, Nanna. 1996. Fugenelemente. In *Deutsch-typologisch*, 525–550. de Gruyter.
- Goldsmith, John Anton. 1976. *Autosegmental phonology*. Cambridge, MA: Massachusetts Institute of Technology dissertation.
- Hayes, Bruce. 1989. Compensatory lengthening in moraic phonology. *Linguistic inquiry* 20(2). 253–306.
- Inkelas, Sharon. 1989. *Prosodic constituency in the lexicon*. Stanford, CA: Stanford University dissertation.
- Itô, Junko & Armin Mester. 2003. *Japanese morphophonemics: markedness and word structure*. Vol. 41. MIT Press.
- Kiparsky, Paul. 1982. Lexical phonology and morphology. In *Linguistics in the morning calm*, 3–91. Seoul: Hansin.
- Kirchner, Jesse Saba. 2009. Kwakwala mu: t reduplication without RED. In *Proceedings of wscla*, vol. 14.
- Kürschner, Sebastian & Renata Szczepaniak. 2013. Linking elements origin, change, and functionalization. *Morphology* 23. 1–6.
- Labrune, Laurence. 1999. Variation intra et inter-langue: morpho-phonologie du rendaku en japonais et du sai-sios en coréen. *Cahiers de grammaire* 24. 117–152.
- Labrune, Laurence. 2014. Featural linking elements in Basque compounds. *Morphology* 24(4). 377–405.
- Legendre, Geraldine, Yoshiro Miyata & Paul Smolensky. 1990. Can Connectionism Contribute to Syntax? Harmonic Grammar, with an Application. *CLS* 26(1). 237–252.
- Mohanan, K. P. 1982. *Lexical phonology*. MA: Cambridge, Massachusetts Institute of Technology dissertation.
- Mohanan, K. P. 1986. *The theory of lexical phonology*. Dordrecht: D. Reidel Publishing Company.

- Mohanan, Tara. 1989. Syllable structure in Malayalam. *Linguistic Inquiry*. 589–625.
- Namboodiripad, Savithry, Eric Bakovic & Marc Garellek. 2015. Moraic geminates in Malayalam: Evidence from minimal word effects and loanword adaptation.
- Nóbrega, Vitor A. 2020. No escape from categorization: an insiders view of compounds. *Ilha do Desterro* 73(3). 103–126.
- Potts, Christopher, Joe Pater, Karen Jesney, Rajesh Bhatt & Michael Becker. 2010. Harmonic Grammar with linear programming: from linear systems to linguistic typology. *Phonology* 27(1). 77–117.
- Ralli, Angela. 2008. Compound markers and parametric variation. *STUF - Language Typology and Universals* 61(1). 19–38.
- Ralli, Angela. 2019. Coordination in compounds. In *Oxford research encyclopedia of linguistics*. Oxford University Press Online. accessed 26. September 2024. <https://doi.org/https://doi.org/10.1093/acrefore/9780199384655.013.564>.
- Rosen, Eric. 2016. Predicting the unpredictable: Capturing the apparent semi-regularity of rendaku voicing in Japanese through harmonic grammar. In *Proceedings of bls*, vol. 42, 235–249.
- Selkirk, Elisabeth O. 2011. The syntax-phonology interface. In John Goldsmith, Jason Riggle & Alan C. L. Yu (eds.), *The handbook of Phonological Theory*, 435–484. Wiley Online Library.
- Smolensky, Paul & Matthew Goldrick. 2016. Gradient symbolic representations in grammar: The case of French liaison. *Rutgers Optimality Archive* 1552. 1–37.
- Sproat, Richard. 1985. *On deriving the lexicon*. Cambridge, MA: MIT dissertation.
- Trommer, Jochen & Eva Zimmermann. 2014. Generalised mora affixation and quantity-manipulating morphology. *Phonology* 31(3). 463–510.
- Wälchli, Bernhard. 2005. *Co-compounds and Natural Coordination*. Oxford: Oxford University Press.
- Zimmermann, Eva. 2018. Gradient symbolic representations in the output: A case study from MosesColumbian Salishan stress. *Proceedings of NELS* 48. 275–284.
- Zimmermann, Eva. 2019. Gradient symbolic representations and the typology of ghost segments. In *Proceedings of the annual meetings on phonology*, vol. 6.